



Pakistan Institute of Engineering and Applied Sciences (PIEAS)

Department of Electrical Engineering

Circuit Analysis-I

Complex Engineering Problem

House Wiring

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1 Introduction

1.1 Objectives

The objective of this project is to convert a 5-marla house layout into a complex engineering problem by designing its complete electrical circuit diagram and load distribution using Multisim and MATLAB.

1.2 Abstract

This project designs and analyzes the electrical wiring of a normal house with three bedrooms, a kitchen, a living room, and a guest room. A complete circuit diagram is created using circuit-making software, showing all loads, switches, MCBs, and a separate UPS line for appliances below 100 W. MATLAB is used to calculate current, voltage, and power for each appliance, each room, and the overall home for both main supply and UPS operation. The circuit is also simulated in Multisim/LTspice to verify connections, observe circuit behavior, and detect possible overloads or faults. Suitable ratings for wires, MCBs, and the UPS are selected according to the calculated load requirements. The project also checks load balancing across rooms to avoid overloading any part of the circuit. Energy efficiency and safety considerations are included while selecting components. The UPS performance is compared with the main supply to understand backup limitations. Overall, this project gives a clear understanding of practical home wiring, safe load distribution, and the role of a UPS in keeping essential appliances running during power outages.

1.3 Resources

- **MULTISIM:** Used to design the complete house circuit layout with all loads and connections. It simulates how the circuit works in real conditions. It also helps check faults, current flow, and proper operation of components.
- **MATLAB:** Used to calculate power, current, voltage, and resistance for all appliances and rooms. It helps compare main supply and UPS load values. It also supports selecting the correct wire sizes, MCB ratings, and UPS capacity.

2 Analytical Approach

Here is the same content rewritten in a simple, clear paragraph with no formulas:

The house wiring in this project is designed in a parallel configuration with the main 240 V supply so that every room receives full voltage independently and continues working even if a fault occurs in one area. The model includes three bedrooms, a kitchen, a living room, a guest room, and the porch and backyard circuits. All rooms are connected in parallel because this method keeps each room separate and prevents the entire system from shutting down if one appliance fails, unlike a series circuit. To keep the analysis simple, a DC source is used instead of AC, avoiding phase calculations that change in real life. Every appliance in the house is represented by a resistance value based on its power rating, using realistic values commonly found in the market. Capacitors and inductors, although present in actual wiring, were removed from the design to meet the requirement of creating a simple home circuit. Current for each device and room was calculated, and the total power supplied and consumed was compared. This approach makes it easy to understand load distribution, safe wiring design, and how the main supply and UPS support the house under different conditions.

3 Home Circuit Layout

The following rooms and areas are included:

- Bedrooms (3)
- Guest Room
- Bathrooms (2)
- Kitchen
- Living Room
- Study Room
- Garage
- Patio
- Store Room

3.1 House sketch

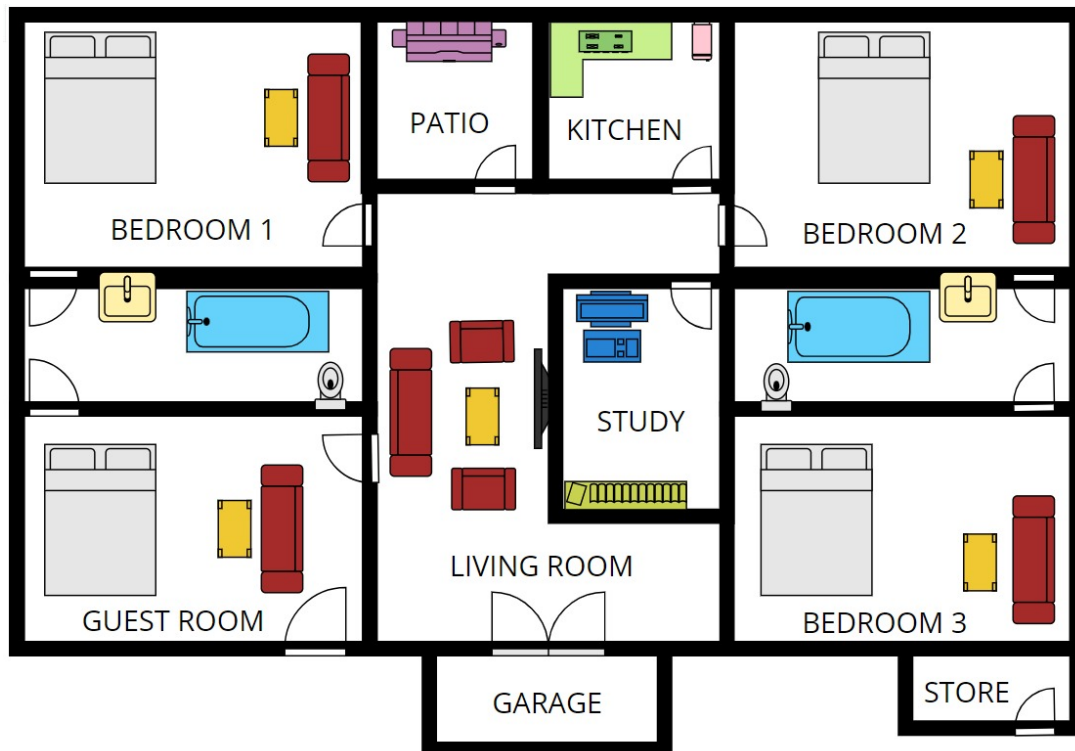
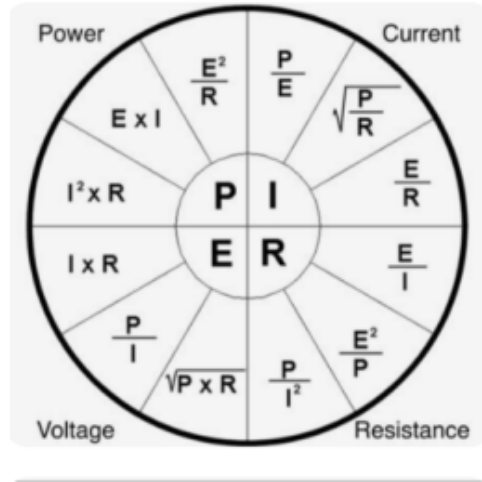


Figure 1: Basic House sketch

The floor plan shows a single-story house with a clear and organized layout based on its electrical schematic. It includes three bedrooms, two bathrooms, a central living room, a kitchen, a guest room, a laundry area, and a garage. The living room is centrally located for easy access to all areas. Bedroom 1 and Bathroom 1 are on the lower left, while Bedroom 2 and Bathroom 2 are slightly toward the top left. Bedroom 3 is next to the living room, opposite the kitchen. The guest room and garage are on the right side, with the laundry area near the garage entry. Doors are marked to show swing directions, and the layout groups rooms logically, reflecting the house's electrical zones.



4 Electrical Calculations for Household Appliances

The electrical parameters for selected household appliances were calculated using MATLAB. Resistance and current values were determined based on rated power with a supply voltage of 240 V.

Formulae used for calculations:

- $R = \frac{V^2}{P}$ (Resistance in ohms)
- $I = \frac{V}{R}$ (Current in amperes)

Values calculated for the appliances with the given power rating (using MATLAB):

Appliance	Power Rating (W)	Resistance (Ω)	Current (A)
LED Light	15.0	3840.0	0.0625
Ceiling Fan	60.0	960.0	0.2500
AC (1.0 TON)	1000.0	57.6	4.1667
Lamp	60.0	960.0	0.2500
Computer	200.0	288.0	0.8333
Exhaust Fan	90.0	640.0	0.3750
Refrigerator	150.0	384.0	0.6250
Microwave Oven	1200.0	48.0	5.0000
Washing Machine	750.0	76.8	3.1250
Electric Iron	750.0	76.8	3.1250
Water Pump	750.0	76.8	3.1250
Chandelier	60.0	960.0	0.2500
LCD TV	150.0	384.0	0.6250

Table 1: Electrical calculations of selected household appliances at 240 V supply.

4.1 Matlab Calculations For HouseHold Appliances :

4.1.1 For Exhaust and Ac :

```
>> V = 240;  
P_EXHAUST = 90;  
R_EXHAUST = V*V / P_EXHAUST  
I_EXHAUST = P_EXHAUST / V  
  
R_EXHAUST =  
  
640  
  
I_EXHAUST =  
  
0.3750
```

Figure 2: EXHAUST

```
>> V = 240;  
P_AC = 1000;  
R_AC = V*V / P_AC  
I_AC = P_AC / V  
  
R_AC =  
  
57.6000  
  
I_AC =  
  
4.1667
```

Figure 3: AC

4.1.2 For Chandelier and Iron :

```
>> V = 240;  
P_CHANDELIER = 60;  
R_CHANDELIER = V*V / P_CHANDELIER  
I_CHANDELIER = P_CHANDELIER / V  
  
R_CHANDELIER =  
  
960  
  
I_CHANDELIER =  
  
0.2500
```

Figure 4: CHANDELIER

```
>> V = 240;  
P_IRON = 750;  
R_IRON = V*V / P_IRON  
I_IRON = P_IRON / V  
  
R_IRON =  
  
76.8000  
  
I_IRON =  
  
3.1250
```

Figure 5: IRON

4.1.3 For Washing Machine And Fridge

```
>> V = 240;
P_WashingMachine = 750;
R_WashingMachine = V*V / P_WashingMachine
I_WashingMachine = P_WashingMachine / V

R_WashingMachine =

    76.8000

I_WashingMachine =

    3.1250
|
```

Figure 6: WASHING MACHINE

```
>> V = 240;
P_FRIDGE = 150;
R_FRIDGE = V*V / P_FRIDGE
I_FRIDGE = P_FRIDGE / V

R_FRIDGE =

    384

I_FRIDGE =

    0.6250
```

Figure 7: FRIDGE

4.1.4 For Oven And Computer

```
>> V = 240;
P_OVEN = 1200;
R_OVEN = V*V / P_OVEN
I_OVEN = P_OVEN / V

R_OVEN =

    48

I_OVEN =

    5
```

Figure 8: OVEN

```
>> V = 240;
P_COMPUTER = 200;
R_COMPUTER = V*V / P_COMPUTER
I_COMPUTER = P_COMPUTER / V

R_COMPUTER =

    288

I_COMPUTER =

    0.8333
```

Figure 9: COMPUTER

4.1.5 For WaterPump And Light :

```
>> V = 240;
P_WATERPUMP = 750;
R_WATERPUMP = V*V / P_WATERPUMP
I_WATERPUMP = P_WATERPUMP / V

R_WATERPUMP =

    76.8000

I_WATERPUMP =

    3.1250
|
```

Figure 10: WATERPUMP

```
>> V = 240;
P_LIGHT = 15;
R_LIGHT = V*V / P_LIGHT
I_LIGHT = P_LIGHT / V

R_LIGHT =

    3840

I_LIGHT =

    0.0625
```

Figure 11: LIGHT

4.1.6 For Fan And Lcd :

```
>> V = 240;
P_FAN = 60;
R_FAN = V*V / P_FAN
I_FAN = P_FAN / V

R_FAN =

    960

I_FAN =

    0.2500
```

Figure 12: FAN

```
>> V = 240;
P_LCD = 150;
R_LCD = V*V / P_LCD
I_LCD = P_LCD / V

R_LCD =

    384

I_LCD =

    0.6250
```

Figure 13: LCD

5 Electrical Calculations For House

5.1 Formulas Used

- **Ohm's Law:** $V = IR$
- **Resistance from Power:** $R = \frac{V^2}{P}$
- **Power:** $P = IV$
- **Room Current:** $I = \frac{P}{240 \text{ V}}$

5.2 Room Power and Current Calculations

Table 2: Power and Current per Room (at 240 V)

Room / Area	Appliances	Power (W)	Current (A)
Bedroom 1	AC + Fan + 2 Lights	1090	4.54
Bedroom 2	Fan + 2 Lights	90	0.38
Bedroom 3	Fan + 2 Lights	90	0.38
Guest Room	Fan + 2 Lights	90	0.38
Bathrooms (2)	2 Lights each	60	0.25
Kitchen	2 Lights + Oven + Fridge + Exhaust Fan	1470	6.13
Living Room	3 Lights + AC + Chandelier + LCD TV	1255	5.23
Study Room	2 Lights + Fan + Computer	290	1.21
Garage	2 Lights	30	0.13
Patio / Utility	Lights + Fan + Iron + Washing Machine	1590	6.63
Store Room	2 Lights + Water Pump	780	3.25

5.3 Total Home Power and Current

The total power of the home is calculated by summing the power of all rooms:

$$\begin{aligned} P_{\text{home}} &= P_{\text{Bedroom 1}} + P_{\text{Bedroom 2}} + P_{\text{Bedroom 3}} + P_{\text{Guest Room}} \\ &\quad + P_{\text{Bathrooms}} + P_{\text{Kitchen}} + P_{\text{Living Room}} + P_{\text{Study Room}} \\ &\quad + P_{\text{Garage}} + P_{\text{Patio / Utility}} + P_{\text{Store Room}} \\ &= 1090 + 90 + 90 + 90 + 60 + 1470 + 1255 + 290 + 30 + 1590 + 780 \\ &= 6835 \text{ W} \end{aligned}$$

$$P_{\text{home}} = 6835 \text{ W}$$

Correspondingly, the total home current at 240 V supply is:

$$I_{\text{home}} = \frac{P_{\text{home}}}{240} \approx 28.48 \text{ A}$$

5.4 Matlab Calculations For House

5.4.1 For Bedroom 1 and 2

```
>> P_BEDROOM1 = P_AC + P_FAN + 2*P_LIGHT  
  
P_BEDROOM1 =  
  
1090
```

Figure 14: BEDROOM 1

```
>> P_BEDROOM2 = P_FAN + 2*P_LIGHT  
  
P_BEDROOM2 =  
  
90
```

Figure 15: BEDROOM 2

5.4.2 For Bedroom 3 and Guest Room

```
>> P_BEDROOM3 = P_FAN + 2*P_LIGHT  
  
P_BEDROOM3 =  
  
90
```

Figure 16: BEDROOM 3

```
>> P_GUEST = P_FAN + 2*P_LIGHT  
  
P_GUEST =  
  
90
```

Figure 17: GUEST ROOM

5.4.3 For Washrooms And Kitchen

```
>> P_WASHROOMS = 4 * P_LIGHT  
  
P_WASHROOMS =  
  
60
```

Figure 18: WASHROOMS

```
>> P_KITCHEN = 2*P_LIGHT + P_OVEN + P_FRIDGE + P_EXHAUST  
  
P_KITCHEN =  
  
1470
```

Figure 19: KITCHEN

5.4.4 For Study And Living Room

```
>> P_LIVING = 3*P_LIGHT + P_AC + P_CHANDELIER + P_LCD  
  
P_LIVING =  
  
1255
```

Figure 20: LIVING ROOM

```
>> P_STUDY = 2*P_LIGHT + P_FAN + P_COMPUTER  
  
P_STUDY =  
  
290
```

Figure 21: STUDY ROOM

5.4.5 For Garage And Patio

$$\begin{aligned} P_{\text{GARAGE}} &= 2 * P_{\text{LIGHT}} \\ P_{\text{GARAGE}} &= \\ 30 \end{aligned}$$

Figure 22: GARAGE

$$\begin{aligned} P_{\text{PATIO}} &= 2 * P_{\text{LIGHT}} + P_{\text{FAN}} + P_{\text{IRON}} + P_{\text{WASHING_MACHINE}} \\ P_{\text{PATIO}} &= \\ 1590 \end{aligned}$$

Figure 23: PATIO

5.4.6 For Store Room

$$\begin{aligned} P_{\text{STORE}} &= 2 * P_{\text{LIGHT}} + P_{\text{WATERPUMP}} \\ P_{\text{STORE}} &= \\ 780 \end{aligned}$$

Figure 24: STORE ROOM

5.5 TOTAL HOUSE POWER ABSORBED :

$$\begin{aligned} P_{\text{HOME}} &= P_{\text{BEDROOM1}} + P_{\text{BEDROOM2}} + P_{\text{BEDROOM3}} + P_{\text{GUEST}} + \dots \\ &\quad P_{\text{WASHROOMS}} + P_{\text{KITCHEN}} + P_{\text{LIVING}} + P_{\text{STUDY}} + \dots \\ &\quad P_{\text{GARAGE}} + P_{\text{PATIO}} + P_{\text{STORE}} \\ P_{\text{HOME}} &= \\ 6835 \end{aligned}$$

Figure 25: HOUSE POWER

6 HOUSE CIRCUIT DIAGRAM :

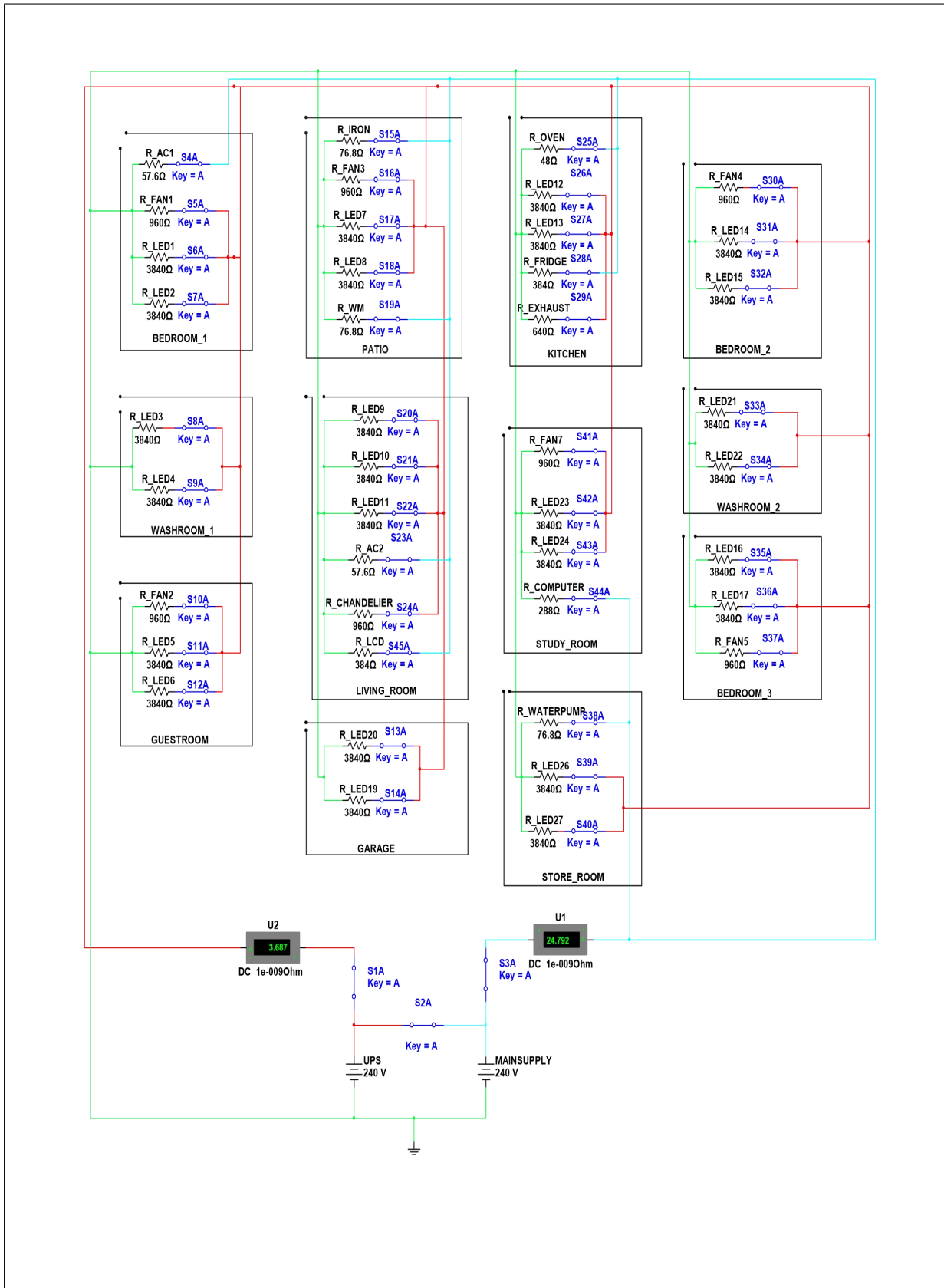
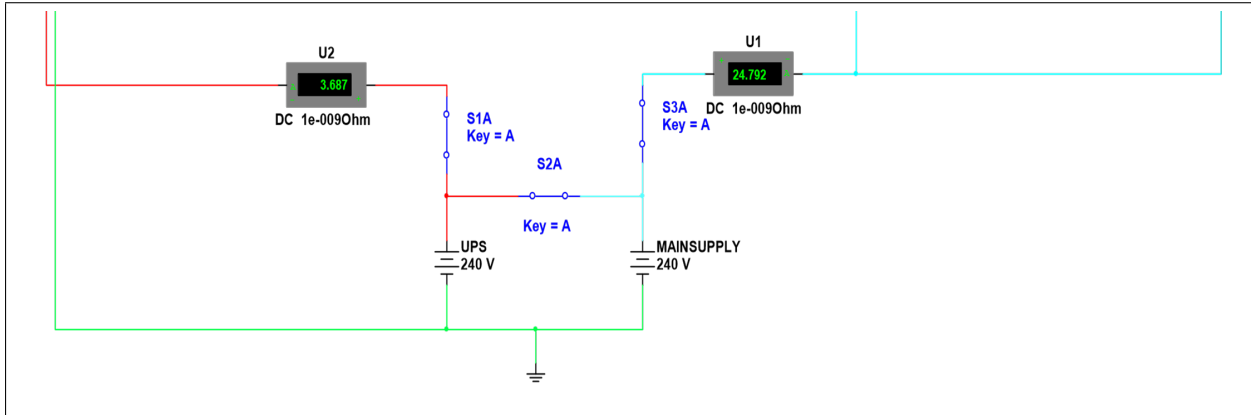


Figure 26: CIRCUIT DIAGRAM

7 PROVEN STATEMENT FOR POWER DELIVERED AND ABSORBED

subsectionPOWER DELIVERED :



7.0.1 POWER DELIVERED CALCULATIONS :

Given:

$$V = 240 \text{ V}, \quad U_1 = 24.792 \text{ A}, \quad U_2 = 3.687 \text{ A}$$

Current:

$$I = U_1 + U_2 = 24.792 + 3.687 = 28.479 \text{ A}$$

Power delivered:

$$P_{\text{delivered}} = V \cdot I = 240 \cdot 28.479 \approx 6835 \text{ W}$$

7.1 POWER ABSORBED :

$$\begin{aligned}
 P_{\text{HOME}} &= P_{\text{BEDROOM1}} + P_{\text{BEDROOM2}} + P_{\text{BEDROOM3}} + P_{\text{GUEST}} + \dots \\
 &\quad P_{\text{WASHROOMS}} + P_{\text{KITCHEN}} + P_{\text{LIVING}} + P_{\text{STUDY}} + \dots \\
 &\quad P_{\text{GARAGE}} + P_{\text{PATIO}} + P_{\text{STORE}} \\
 P_{\text{HOME}} &= \\
 &\quad 6835
 \end{aligned}$$

Figure 27: HOUSE POWER

7.2 The power delivered is equal to the power absorbed:

$$P_{\text{delivered}} = P_{\text{absorbed}}$$

Errors and Uncertainties

In this complex engineering problem, we assumed ideal devices and resistors, which are not practically realizable. The following are the key sources of errors and uncertainties in our analysis:

1. We used an ideal DC current source, which is not practically available. In real-world household circuits, an AC current source is used since the voltage varies continuously, and household appliances are designed to tolerate these variations.
2. Our home circuit also assumes an ideal DC voltage source, with a constant voltage throughout, which does not reflect the fluctuating nature of real AC voltage.
3. The internal resistance of wires was ignored, which affects the power ratings of connected appliances. Instead, we used simplified power calculations based on Ohm's Law.
4. We assumed a constant 240 V supply, whereas in reality, the AC voltage continuously changes in magnitude in a standard home wiring system.
5. Using a DC source means we could not incorporate the frequency aspect of AC, which in Pakistan is 50 Hz. Frequency plays a crucial role in real household circuits.
6. Transformers and plugs, essential for high-power appliances such as air conditioners and refrigerators, could not be used in our DC-based analysis. Consequently, power ratings were calculated using simple Ohm's Law, which deviates from practical considerations.
7. Appliance resistances were assumed ideal, without considering their tolerances, which further deviates from practical circuit behavior.

Overall, while our home circuit provides a simplified example of a standard household setup, these approximations and uncertainties make it different from real-world household circuits.

Conclusion

Despite these limitations, this complex engineering problem offers several valuable learning outcomes:

1. It enhanced our practical skills and our ability to use technical software such as MATLAB and MULTISIM for both circuit construction and analysis of currents and power ratings.
2. We learned how theoretical concepts can be translated into practical implementations, bridging the gap between calculations and real-world applications.
3. By designing the home circuit, we practiced connecting resistors and appliances in series, integrating voltage sources appropriately, grounding circuits to avoid open circuits, and terminating connections correctly.
4. We reinforced our understanding of fundamental electrical laws, such as Ohm's Law, and their application in practical circuit design.

In conclusion, this complex engineering problem successfully linked theoretical knowledge with practical skills, improving both our understanding of electrical principles and our hands-on circuit design capabilities.